

Erdős Problem #162

EP-162	L1 - Tractable	Partiellement résolu	Graph Theory
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Énoncé original (anglais, reproduction mot pour mot)

Let $\alpha > 0$ and $n \geq 1$. Let $F(n, \alpha)$ be the largest k such that there exists some 2-colouring of the edges of K_n in which any induced subgraph H on at least k vertices contains more than $\alpha \binom{|V(H)|}{2}$ many edges of each colour.

Prove that for every fixed $0 \leq \alpha \leq 1/2$, as $n \rightarrow \infty$, $F(n, \alpha) \sim c_\alpha \log n$ for some constant c_α .

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